

Postpartum Hemorrhage 03: Uterotonics, TXA, and Medication Choices

Companion reading handout for simultaneous listening.

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Uterotonics, tranexamic acid, and medication choices.

In the last episode, we built the first five minutes.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

Now we slow down and examine the medication track

This episode is about drugs, but it is not really about memorizing a table.

A drug table is useful when the room is quiet.

In a hemorrhage, you need something faster.

You need a clinical reflex.

So here is the central rule.

Uterotonics treat atony.

They do not treat all postpartum hemorrhage.

Say that again.

Uterotonics treat atony.

They do not repair a cervical laceration.

They do not remove retained placenta.

They do not reverse coagulopathy.

They do not close a uterine artery.

If the uterus is soft, uterotonics are central.

If the uterus is firm and bleeding continues, uterotonics may be background support, but the main problem is probably somewhere else.

This distinction is where many PPH responses either become sharp or become repetitive.

Sharp response asks:

Is this tone?

Repetitive response keeps giving one more medication while the real source is trauma, tissue, or thrombin.

The anchor for today is simple.

First, give oxytocin.

Second, add tranexamic acid early.

Third, choose the next uterotonic by contraindication, not by habit.

Fourth, if two uterotonics have not controlled ongoing bleeding, stop treating the situation as only a medication problem.

Let us start with oxytocin

Oxytocin is first-line for PPH caused by uterine atony.

It is first-line because it is effective, familiar, and has a relatively favorable side effect profile.

The local protocol gives oxytocin as 10 units intramuscularly, or 10 to 40 units in 500 to 1000 milliliters of normal saline or Hartmann's solution by intravenous infusion.

The Gabbe chapter describes 10 to 80 units in 500 to 1000 milliliters of crystalloid, with higher-dose infusion shown to reduce the need for additional hemorrhage treatment in some data.

For the resident at the bedside, the practical point is this:

Oxytocin is your first uterotonic, and infusion is preferred for treatment.

Do not give it as a rapid intravenous push.

A rapid push can cause hypotension.

That is the exact opposite of what you want in a hemorrhaging patient.

Oxytocin also has an anti-diuretic effect because of structural similarity to anti-diuretic hormone. With prolonged excessive dosing, especially in hypotonic fluids, water intoxication and hyponatremia can occur.

That does not make oxytocin dangerous in routine PPH treatment.

It means you should respect the dose, the fluid, and the duration.

So the oxytocin script is:

Soft uterus. Start oxytocin. Use infusion. Avoid rapid push. Keep watching the clinical response.

What response are you watching?

You are watching the uterus and the bleeding.

Does the uterus firm?

Does the bleeding slow?

Does the fundus stay contracted when massage relaxes?

Does the blood loss continue to climb despite treatment?

The answer determines whether this remains a medication story or becomes an escalation story.

Now let us add tranexamic acid

Tranexamic acid, known locally as Hexakapron, is not a uterotonic.

It does not squeeze the uterus.

It is an antifibrinolytic.

It helps stabilize clot by reducing fibrinolysis.

The position paper includes tranexamic acid 1000 milligrams per dose among first-line treatment considerations.

The local protocol adds Hexakapron 1 gram in addition to uterotonic treatment, and the checklist uses 1 gram in 100 milliliters.

For audio memory, keep it simple:

Tranexamic acid is 1 gram intravenously.

It is an adjunct.

It does not replace source control.

That last sentence matters.

Tranexamic acid can support hemostasis, but it cannot repair a tear. It cannot empty a uterus full of retained tissue. It cannot tamponade a placental bed that is pouring blood. It cannot substitute for a Bakri balloon, embolization, laparotomy, or hysterectomy when those are needed.

So the tranexamic acid script is:

Add tranexamic acid early. One gram intravenously. Then keep looking for the source.

Now we come to the second-line uterotonics

This is where the table becomes a game of exclusions.

Do not ask first, "What do I usually give next?"

Ask first, "What must I not give this patient?"

Hypertensive?

Methergine is out.

Preeclampsia?

Methergine is out.

Asthma?

Carboprost is out, or at minimum you should treat it as a major contraindication and avoid it in the acute setting.

Active pulmonary disease?

Carboprost is out.

Active liver, renal, or cardiac disease?

Be very cautious with carboprost, and follow the protocol contraindications.

That is the bedside game.

Hypertensive: no Methergine.

Asthmatic: no carboprost.

Micrograms for misoprostol, never milligrams.

Persistent bleeding after two uterotonics: escalate.

Let us take them one by one.

Methylergometrine, or Methergine, is an ergot alkaloid that increases and synchronizes uterine contraction.

The local protocol uses 0.2 milligrams intramuscularly, repeated every 2 to 4 hours if needed.

Intramuscular use is preferred for early PPH treatment.

The major trap is hypertension.

Methergine can cause severe hypertension. It can also be associated with headache, blurred vision, seizure, syncope, chest pain, and even myocardial infarction in severe cases.

The local protocol says intravenous use should be avoided because of sudden hypertension and even stroke. If intravenous use is life-savingly necessary, it must be slow, over at least 60 seconds, with strict blood pressure monitoring during and after administration.

The absolute contraindications locally are hypertension, preeclampsia, and known allergy.

The protocol also advises caution in sepsis, kidney disease, liver disease, and use of certain strong CYP3A4 inhibitors, such as macrolides, HIV medications, and antifungal drugs.

So here is the Methergine vignette.

A woman has atony after delivery. Her uterus is boggy. Bleeding is ongoing. Her blood pressure is 160 over 105, and she has preeclampsia.

Do not give Methergine.

This is the classic trap.

You may feel pressure to give "the next uterotonic," but the next uterotonic cannot be chosen from habit. It must be chosen from the patient's physiology.

The Methergine script is:

Methergine is 0.2 milligrams intramuscularly. It is useful for atony, but not in hypertension or preeclampsia.

Now carboprost

Carboprost is prostaglandin F two alpha. It is also known as Hemabate.

It is a powerful uterotonic for atony that does not respond adequately to oxytocin and other conventional measures.

The local protocol uses 0.25 milligrams intramuscularly.

It may be repeated every 15 to 90 minutes according to clinical judgment, with no more than 8 doses, for a total maximum of 2 milligrams.

The checklist writes 250 micrograms, which is the same as 0.25 milligrams.

The key side effects come from smooth muscle activity: nausea, vomiting, diarrhea, fever, chills, headache, hypertension, and bronchospasm.

The key contraindication is asthma or active lung disease.

Gabbe emphasizes asthma as a strong contraindication because prostaglandin F two alpha has bronchoconstrictive properties.

The local protocol also lists known allergy, chorioamnionitis, known liver or kidney disease, active cardiovascular disease, and active obstructive pulmonary disease as contraindications, and notes hypertension as a relative contraindication in the table.

So here is the carboprost vignette.

A woman has uterine atony and ongoing bleeding.

She also has asthma.

Do not reach automatically for carboprost.

In the hemorrhage room, asthma is not a small footnote. It is the warning sign on the drug.

The carboprost script is:

Carboprost is 0.25 milligrams intramuscularly, repeat every 15 to 90 minutes, maximum 8 doses. Avoid it in asthma and active pulmonary disease.

Now misoprostol

Misoprostol, or Cytotec, is a prostaglandin E one analogue.

It is useful because it is inexpensive, effective, stable, and does not require refrigeration.

It can be given through multiple routes.

Gabbe lists 600 to 1000 micrograms.

The local checklist gives options: 800 to 1000 micrograms rectally, 600 micrograms orally, or 800 micrograms sublingually.

The departmental table appears to write 600 to 1000 milligrams in one place.

That is not the dose you should teach.

That is a unit error.

Misoprostol for PPH is taught in micrograms, not milligrams.

Micrograms, never milligrams.

Say it in your head in the drug round:

Miso is micrograms.

Common side effects include nausea, vomiting, diarrhea, fever, chills, and headache.

There are few absolute contraindications in the PPH context, aside from rare prostaglandin sensitivity in the local protocol.

But the important bedside point is not only safety.

The important point is not to let misoprostol become the third hopeful medication while the patient is crossing into Stage two.

If atony persists and bleeding continues after two uterotonics, the next move may be advanced control, not a longer medication sequence.

That means the operating room, Bakri balloon after excluding trauma, embolization if stable and feasible, or surgical treatment.

So the misoprostol script is:

Misoprostol is 600 to 1000 micrograms, route-dependent. It is useful, stable, and flexible. Remember micrograms, not milligrams.

Now calcium gluconate

This is not one of the classic uterotonic table drugs, but it is a useful pearl from Gabbe.

Sometimes atony is worsened by medications that impair calcium entry into the myometrial cell.

The examples are magnesium sulfate and nifedipine.

If uterine atony is related to those drugs, calcium gluconate can be considered as an adjuvant.

Gabbe describes 1 gram in 10 milliliters as an intravenous push that can improve uterine tone in that setting.

This is not a replacement for oxytocin, uterine massage, tranexamic acid, source control, or escalation.

It is a targeted adjunct when the mechanism fits.

Here is the vignette.

A patient with severe preeclampsia is receiving magnesium sulfate.

She delivers and develops atony.

Her uterus is poorly contracted.

Oxytocin is running.

Methergine is contraindicated because of hypertension and preeclampsia.

Carboprost may be considered if she has no asthma or pulmonary contraindication, but you also remember that magnesium can impair contraction through calcium physiology.

This is a moment to consider calcium gluconate as an adjunct.

That kind of thinking is what separates protocol memory from clinical understanding.

Now let us put the medication choices into a bedside sequence

The uterus is soft.

You are massaging.

The bladder is being emptied.

Large-bore access is being placed.

Blood is being sent.

Help has been called.

You start oxytocin.

You add tranexamic acid early.

Now the uterus is still soft.

You choose the next uterotonic.

If the patient is hypertensive or preeclamptic, avoid Methergine.

If the patient has asthma or active pulmonary disease, avoid carboprost.

If you use misoprostol, remember micrograms.

If magnesium or nifedipine may be contributing to poor tone, consider calcium gluconate as an adjuvant.

And through all of this, keep asking the harder question:

Is this still tone?

Because medication confidence can become medication tunnel vision.

Let us use four rapid cases.

Case one

Soft uterus. Blood pressure normal. No asthma. Placenta appears complete. Bleeding slows with massage but resumes when the uterus relaxes.

This is a straightforward atony pattern.

Oxytocin is first.

Tranexamic acid is added.

Methergine can be used if no contraindication.

Carboprost can be used if no pulmonary contraindication.

Misoprostol is another option.

But if bleeding continues after two uterotonics, you are not proving that the third drug is magic. You are proving that escalation must begin.

Case two

Soft uterus. Severe preeclampsia. Blood pressure 165 over 110.

The uterus needs treatment, but Methergine is not the treatment.

Use oxytocin.

Add tranexamic acid.

Choose other options according to contraindications.

Think of calcium gluconate if magnesium is contributing to atony.

Do not let the stress of bleeding push you into giving a drug that worsens hypertension.

Case three

Soft uterus. History of asthma.

Carboprost is the trap.

Use oxytocin.

Add tranexamic acid.

Consider Methergine if blood pressure and preeclampsia status allow.

Consider misoprostol.

Escalate if the bleeding does not respond.

Case four

The uterus is firm.

Bleeding continues.

This is the case where medication obsession becomes dangerous.

A firm uterus with continuing bleeding is not asking for endless uterotonics.

It is asking you to inspect for trauma, retained tissue, uterine rupture, concealed hematoma, or coagulopathy.

You may continue uterotonic support, but the main action is diagnosis and source control.

Look at the cervix.

Look at the vagina.

Look at the perineum and vulva.

Ask whether the placenta and membranes are complete.

Check coagulation, platelets, and fibrinogen.

Use ultrasound when intra-abdominal bleeding or retained tissue is a concern.

The medication track must never blind you to the four T's.

Now let us handle a subtle but important point about "more"

More medication is not always more treatment.

If the hemorrhage has crossed Stage two thresholds, or if bleeding continues after two uterotonics, the protocol moves toward advanced control.

That may mean operating room transfer.

It may mean Bakri balloon when atony is the cause and tears have been excluded.

It may mean embolization in a stable patient with persistent but slow bleeding when feasible.

It may mean laparotomy, compression sutures, uterine artery ligation, internal iliac ligation, or hysterectomy.

Those are not failures of medical management.

They are the next correct treatments when medication is not enough.

A resident should learn to say this clearly:

We have treated atony medically. Bleeding continues. We are escalating.

That sentence saves time.

It prevents the room from waiting for one more dose while blood loss continues.

Let us close with the memory structure.

Oxytocin first.

Do not push it rapidly intravenously.

Tranexamic acid early.

One gram intravenously.

Methergine is 0.2 milligrams intramuscularly.

No Methergine in hypertension or preeclampsia.

Carboprost is 0.25 milligrams intramuscularly.

No carboprost in asthma or active pulmonary disease.

Misoprostol is micrograms.

Micrograms, never milligrams.

Calcium gluconate is an adjunct when magnesium or nifedipine physiology may be impairing contraction.

And the master rule:

Uterotonics treat tone.

If the uterus is firm, keep looking.

If two uterotonics have not controlled ongoing bleeding, escalate.

In the next episode, we will widen the lens from medications to the four causes of PPH: tone, trauma, tissue, and thrombin. We will learn how to recognize each one at the bedside and how cause-specific thinking prevents the team from treating every hemorrhage as atony.